

Overfitting aur Underfitting (Bias aur Variance)

Model Generalization — Detailed Notes (Hinglish)

1. Dataset ke 3 Types

Jab hum machine learning model banate hain, to hum apna pura data ek hi jagah use nahi karte. Data ko 3 alag parts me baanta jaata hai, taaki hum sahi tarike se model ko train aur evaluate kar sakein:

- **1. Training Dataset** — is data par model ko actually train (seekhna/fit) kiya jaata hai. Model isi data se patterns seekhta hai.
- **2. Test Dataset** — ye data model ne kabhi dekha hi nahi hota. Isko sirf final me use karte hain ye check karne ke liye ki model naye/unseen data par kitna accha perform karta hai.
- **3. Validation Dataset** — ye training dataset ka hi ek chota hissa hota hai jise hum model train karte waqt beech-beech me use karte hain — mainly Hyperparameter Tuning (jaise learning rate, tree depth, etc. set karna) ke liye.

Data Splitting kaise hoti hai — Example

Maan lo humare paas ek Dataset hai jisme total 1000 datapoints hain (jaise Size of House, No. of Bedrooms, Price wala data).

Isse hum ek common ratio jaise 70-30 me split karte hain:

- **Training Dataset (700 points)** → ye data 'Model To Train' ke liye use hota hai, yani model isi 700 data points se seekhta hai.
- **Test Dataset (300 points)** → ye data sirf 'Testing Model' ke liye rakha jaata hai — training ke dauran isko touch nahi karte, sirf akhir me model ko judge karne ke liye use karte hain.

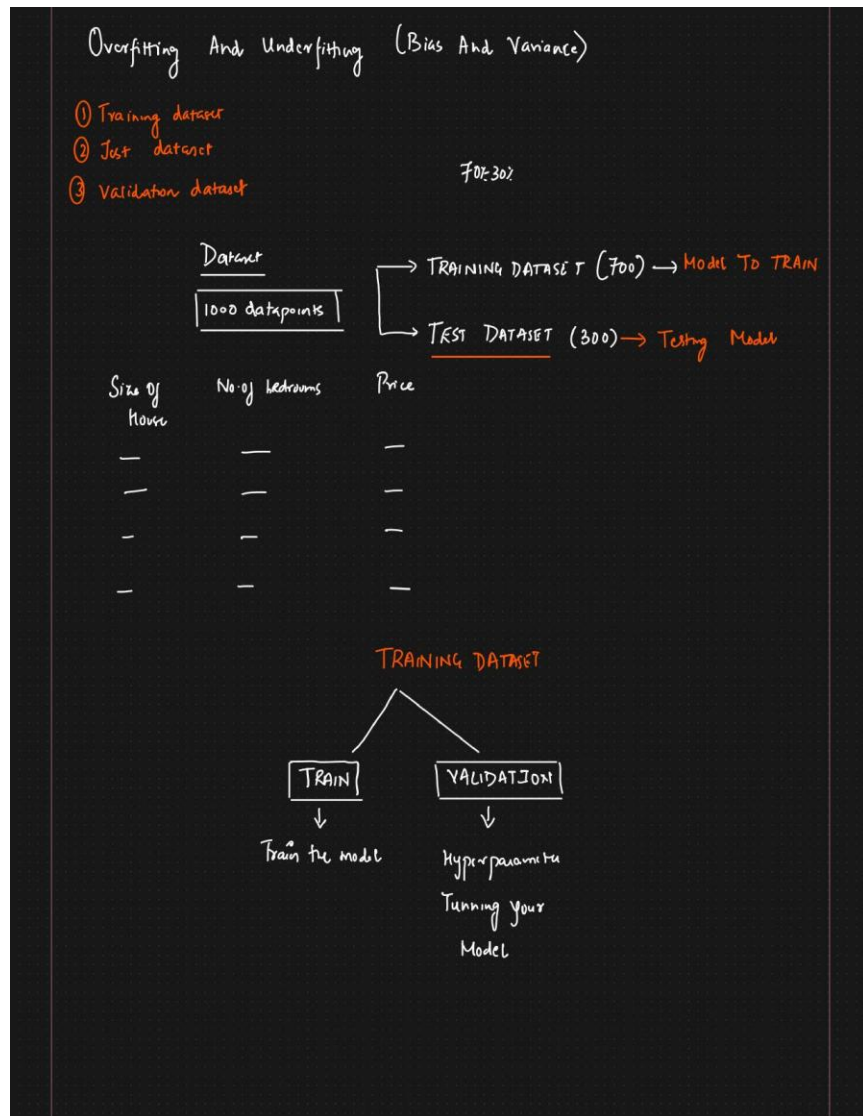
70-30 sirf ek common example ratio hai — practically 80-20 ya 90-10 bhi use hota hai, dataset ke size aur use-case par depend karta hai.

Training Dataset ko aage split karna: Train + Validation

Sirf Training aur Test me split karna kaafi nahi hota — kyunki hume model train karte waqt uske hyperparameters bhi tune (adjust) karne hote hain, aur ye tuning Test data par nahi karni chahiye (warna model 'test data dekh chuka' jaisa ho jaayega, jo galat practice hai).

Isliye Training Dataset ko further do parts me baant dete hain:

- **TRAIN** → is part se model actually seekhta hai (Train the model).
- **VALIDATION** → is part ka use Hyperparameter Tuning ke liye hota hai — yani model ki settings (jaise kitni depth, kitne layers, kitna learning rate) ko best banane ke liye is data par check karte hain, bina Test data ko touch kiye.



Original Notes — Page 1: Dataset Splitting (Training / Test / Validation)

2. Train aur Test Accuracy se Model ko Judge Karna

Model achha hai ya nahi, ye judge karne ka sabse simple tarika hai — Training data par uski accuracy dekho, aur Test (unseen) data par bhi accuracy dekho, phir dono ko compare karo. Isi comparison se pata chalta hai model 'Generalized' hai, 'Overfitting' hai, ya 'Underfitting' hai.

Case 1: Generalized Model (Best Case)

TRAIN Accuracy = 90% (Low Bias) TEST Accuracy = 85% (Low Variance)

Jab Train aur Test — dono par accuracy achhi ho aur dono ke beech ka gap bahut kam ho (jaise 90% aur 85%), to iska matlab model ne data ke real patterns seekhe hain, na ki sirf training data ko 'yaad' kiya hai. Aise model ko Generalized Model kehte hain — yani ye naye/unseen data par bhi bharosemand tarike se kaam karega.

- **Low Bias** — Training data par model ka error kam hai, matlab model ne data ke patterns achhe se seekhe hain.
- **Low Variance** — Train aur Test accuracy ke beech zyada farak nahi hai, matlab model naye data par bhi stable/consistent perform karta hai.

Case 2: Overfitting

TRAIN Accuracy = Very Good (Low Bias) TEST Accuracy = Bad, 50% (High Variance)

Jab model Training data par to bahut achhi accuracy deta hai, lekin Test (naye) data par uski accuracy achanak bahut gir jaati hai (jaise 90% se 50%), to iska matlab model ne training data ko 'ratta' maar liya hai (including uska noise/randomness bhi) — usne general pattern nahi, balki training data ke chhote-chhote details bhi yaad kar liye hain. Isse Test data par model confuse ho jaata hai. Isi condition ko Overfitting kehte hain.

- **Low Bias, High Variance** → yehi Overfitting ki pehchaan hai. Model training data ke saath bahut zyada 'fit' ho jaata hai, lekin naye data ke liye flexible/general nahi reh paata.

Case 3: Underfitting

TRAIN Accuracy = Low (High Bias) TEST Accuracy = Low (High Variance)

Jab model Training data par hi achhi accuracy nahi de pa raha (yani wo apne hi seekhne wale data ke patterns thik se samajh nahi paya), to obviously Test data par bhi wo achha perform nahi karega. Isi situation ko Underfitting kehte hain — model bahut zyada simple hai ya usne data se patterns properly seekhe hi nahi.

- **High Bias** → model training data par bhi galtiyan kar raha hai, matlab usne pattern sahi se seekha hi nahi (model bahut simple hai data ke liye).
- **High Variance (yaha)** → is context me iska matlab hai ki model kisi bhi data (train ho ya test) par reliable achha result nahi de pa raha.

3. Bias aur Variance — Term by Term Samjho

- **Bias** = model ki training data ke actual pattern ko samajhne me jo error/galti hoti hai. Simple shabdon me — model training data par khud kitna galat predict kar raha hai.
- **High Bias** → model training data ko bhi achhe se seekh nahi paya (Underfitting ka sign).
- **Low Bias** → model training data ke pattern ko achhe se seekh gaya.
- **Variance** = model ki performance me kitna 'utaar-chadhaav' (fluctuation) hota hai jab use different datasets (jaise Train vs Test) par try karte hain. Simple shabdon me — model Train aur Test par kitna alag-alag behave kar raha hai.
- **High Variance** → Train aur Test accuracy me bada gap hai (Overfitting ka sign) — model naye data ke sath sensitive/unstable hai.
- **Low Variance** → Train aur Test accuracy lagbhag same hai — model stable hai.

Yaad rakhne ka tarika: 'Bias' = training data par khud ki galti. 'Variance' = naye (unseen) data aane par performance kitni badal jaati hai.

4. Graph se Visually Samjho

Do graphs diye gaye hain — dono me x-axis aur y-axis par data points (X marks) hain, aur ek line/curve fit ki gayi hai jo model ke predictions ko dikhati hai.

Graph 1: Generalized Model

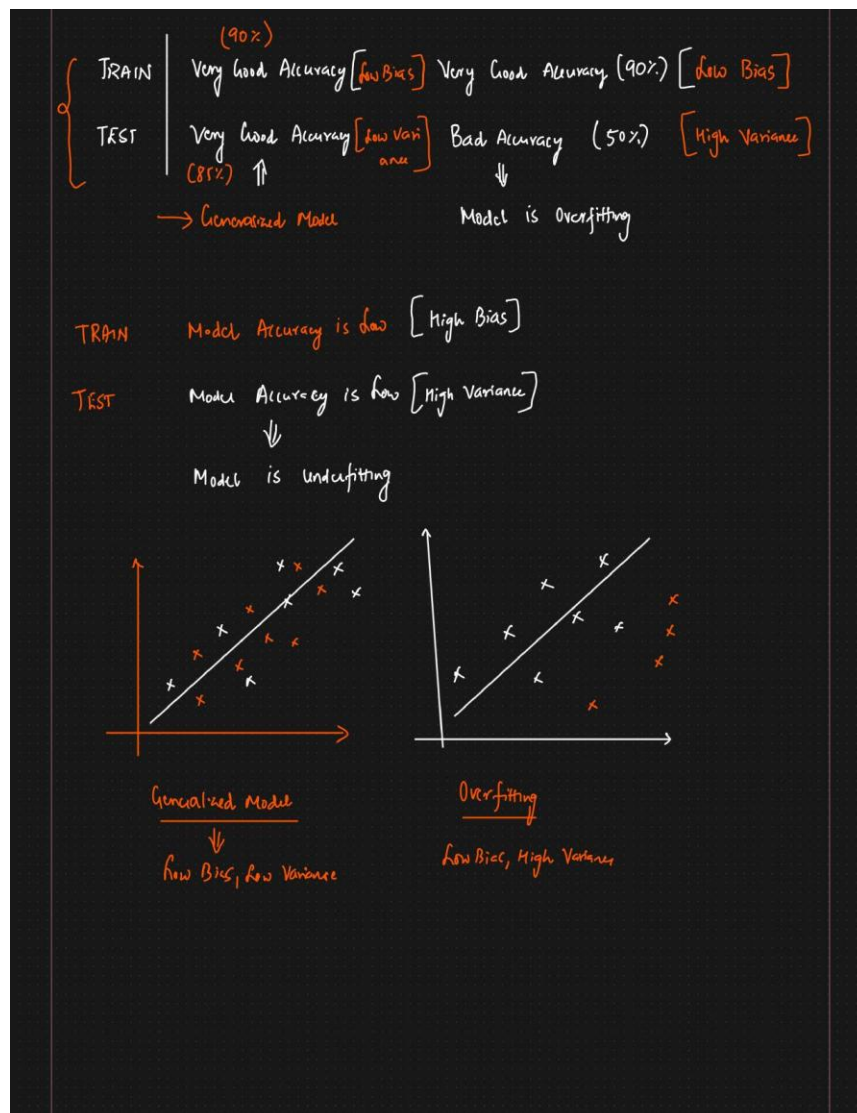
Isme ek seedhi (straight) line data points ke beech se is tarah guzarti hai ki wo overall trend ko capture karti hai, bina har ek chhote point ko exactly touch kiye. Kuch points line se upar hain, kuch neeche — ye normal hai. Yehi ek achha, balanced model hai.

- **Result:** Low Bias, Low Variance — yehi ideal / best situation hai jo hum chahte hain.

Graph 2: Overfitting

Isme line (ya curve) har ek data point ko bahut closely follow karne ki koshish karti hai — jaise wo training ke sab points ko exactly touch karna chahti ho. Aisi line bahut complex/wiggly ban jaati hai. Ye training data par to perfect dikhti hai, lekin naye data point aane par ye galat prediction degi kyunki isne 'noise' ko bhi pattern samajh liya hai.

- **Result:** Low Bias (training par bahut achhi), High Variance (test/naye data par kharab) — yehi Overfitting hai.



Original Notes — Page 2: Train/Test Accuracy Cases, Bias-Variance, Generalized vs Overfitting Graphs

5. Quick Comparison Table

Situation	Train Accuracy	Test Accuracy	Bias / Variance
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Generalized Model	Achhi (~90%)	Achhi, kareeb Train jaisi (~85%)	Low Bias, Low Variance
Overfitting	Bahut Achhi	Kharab (jaise ~50%)	Low Bias, High Variance
Underfitting	Kharab / Low	Kharab / Low	High Bias, High Variance

Overfitting/Underfitting rokne ke common tarike (extra context):

- **Overfitting kam karne ke liye:** zyada data lena, model ko simple banana, Regularization use karna, Cross-Validation karna.
- **Underfitting kam karne ke liye:** model ki complexity badhana, better/relevant features add karna, zyada der train karna.

Ye extra points notes me directly nahi likhe the, lekin concept ko poora samajhne ke liye add kiye gaye hain.